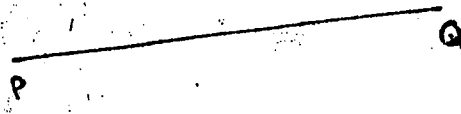


(Exercise 8.2)

15

In each of the problems 1-4, find the parametric eqs., direction ratios, direction cosines & measure of the direction angles of the str. line through P & Q

Q1 $P(1, -2, 0)$ & $Q(5, -10, 1)$

Soln. Given pts. are $P(1, -2, 0)$  Q
& $Q(5, -10, 1)$

Then eq. of line through $P(5, -10, 1)$ & $Q(1, -2, 0)$ is

$$\frac{x-1}{5-1} = \frac{y+2}{-10+2} = \frac{z-0}{1-0}$$

$$\text{or } \frac{x-1}{4} = \frac{y+2}{-8} = \frac{z}{1} = t$$

$$\Rightarrow \left. \begin{aligned} x &= 1+4t \\ y &= -2-8t \\ z &= t \end{aligned} \right\} \text{ are parametric eqs. of given line}$$

D.r.s. of line PQ are $4, -8, 1$

$$\text{Here } \sqrt{4^2 + (-8)^2 + (1)^2} = \sqrt{16 + 64 + 1} = \sqrt{81} = 9$$

So d.c.s. of line through P & Q are

$$\left. \begin{aligned} l &= \frac{4}{9} \\ m &= -\frac{8}{9} \\ n &= \frac{1}{9} \end{aligned} \right\}$$

let α, β, γ be the direction angles of line PQ then¹⁶

$$\left. \begin{aligned} \cos \alpha &= \frac{4}{9} \\ \cos \beta &= -\frac{8}{9} \\ \cos \gamma &= \frac{1}{9} \end{aligned} \right\}$$

$$\Rightarrow \alpha = \cos^{-1}\left(\frac{4}{9}\right) = 63.37^\circ$$

$$\beta = \cos^{-1}\left(-\frac{8}{9}\right) = 152.44^\circ$$

$$\gamma = \cos^{-1}\left(\frac{1}{9}\right) = 83.37^\circ$$

Available at
www.mathcity.org

Q2

$P(6, 5, -3)$ & $Q(4, 1, 1)$

Sol. Given pts. are

$P(6, 5, -3)$ & $Q(4, 1, 1)$

Then eq. of line through $P(6, 5, -3)$ & $Q(4, 1, 1)$ is

$$\frac{x-6}{6-4} = \frac{y-5}{5-1} = \frac{z+3}{-3-1}$$

$$\frac{x-6}{2} = \frac{y-5}{4} = \frac{z+3}{-4}$$

$$\frac{x-6}{1} = \frac{y-5}{2} = \frac{z+3}{-2} = t$$

$$\Rightarrow \left. \begin{aligned} x &= 6+t \\ y &= 5+2t \\ z &= -3-2t \end{aligned} \right\}$$

are parametric eqs. for given line.

Now d.r.s. of line PQ are $1, 2, -2$

$$\text{Now } \sqrt{(1)^2 + (2)^2 + (-2)^2} = \sqrt{1+4+4} = \sqrt{9} = 3$$

So d.c. of line PQ are

$$\left. \begin{aligned} l &= \frac{1}{3} \\ m &= \frac{2}{3} \\ n &= -\frac{2}{3} \end{aligned} \right\}$$

Let α, β, γ be the direction angles of line PQ

Then

$$\left. \begin{aligned} \cos \alpha &= \frac{1}{3} \\ \cos \beta &= \frac{2}{3} \\ \cos \gamma &= -\frac{2}{3} \end{aligned} \right\}$$

$$\Rightarrow \alpha = \cos^{-1}\left(\frac{1}{3}\right) = 70^\circ 32'$$

$$\beta = \cos^{-1}\left(\frac{2}{3}\right) = 48^\circ 11'$$

$$\gamma = \cos^{-1}\left(-\frac{2}{3}\right) = 131^\circ 49'$$

Q3 P(1, -5, 1) & Q(4, -5, 4)

Sol. Given pts. are

P(1, -5, 1) & Q(4, -5, 4)

Then eq. of line through P & Q is

$$\frac{x-1}{4-1} = \frac{y+5}{-5+5} = \frac{z-1}{4-1}$$

$$\frac{x-1}{3} = \frac{y+5}{0} = \frac{z-1}{3}$$

$$\frac{x-1}{1} = \frac{y+5}{0} = \frac{z-1}{1} = t$$

$$\Rightarrow \left. \begin{aligned} x &= 1+t \\ y &= -5+0t \\ z &= 1+t \end{aligned} \right\} \text{ are parametric eq. of line PQ.}$$

Now d.r.s. of PQ are 1, 0, 1

$$\sqrt{(1)^2 + (0)^2 + (1)^2} = \sqrt{1+0+1} = \sqrt{2}$$

So d.c.s. of PQ are

$$\left. \begin{aligned} l &= \frac{1}{\sqrt{2}} \\ m &= 0 \\ n &= \frac{1}{\sqrt{2}} \end{aligned} \right\}$$

Let α, β, γ be the direction angles of line PQ
Then

$$\left. \begin{aligned} \cos \alpha &= \frac{1}{\sqrt{2}} \\ \cos \beta &= 0 \\ \cos \gamma &= \frac{1}{\sqrt{2}} \end{aligned} \right\}$$

$$\Rightarrow \alpha = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

$$\beta = \cos^{-1}(0) = 90^\circ$$

$$\gamma = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right) = 45^\circ$$

Q4 $P(3, 5, 7)$ & $Q(6, -8, 10)$

Soln. Given pts. are $P(3, 5, 7)$ & $Q(6, -8, 10)$.

Then eq. of line PQ is

$$\frac{x-3}{6-3} = \frac{y-5}{-8-5} = \frac{z-7}{10-7}$$

$$\frac{x-3}{3} = \frac{y-5}{-13} = \frac{z-7}{3} = t$$

$$\Rightarrow x = 3 + 3t$$

$$y = 5 - 13t$$

$$z = 7 + 3t$$

are parametric eqs. of given line

Now d.s. of PQ are $3, -13, 3$

$$\sqrt{(3)^2 + (-13)^2 + (3)^2} = \sqrt{9 + 169 + 9} = \sqrt{187}$$

Then the d.s. of line PQ are

$$l = \frac{3}{\sqrt{187}}$$

$$m = -\frac{13}{\sqrt{187}}$$

$$n = \frac{3}{\sqrt{187}}$$

Let α, β, γ be the direction angles of line PQ then

$$\cos \alpha = \frac{3}{\sqrt{187}}$$

$$\cos \beta = -\frac{13}{\sqrt{187}}$$

$$\cos \gamma = \frac{3}{\sqrt{187}}$$

$$\Rightarrow \alpha = \cos^{-1}\left(\frac{3}{\sqrt{187}}\right) = 77.19^\circ$$

$$\beta = \cos^{-1}\left(-\frac{13}{\sqrt{187}}\right) = 159.19^\circ$$

$$\gamma = \cos^{-1}\left(\frac{2}{\sqrt{187}}\right) = 77.19^\circ$$

Q5 Find the Direction Cosines of Co-ord. axes.

Sol. We want to find the d.c.s. of x-axis, y-axis & z-axis.

Since x-axis makes angles $0^\circ, 90^\circ, 90^\circ$ with Co-ord. axes.

So d.c.s. of x-axis are $\cos 0^\circ, \cos 90^\circ, \cos 90^\circ$

$$\text{or } 1, 0, 0$$

Since y-axis makes angles $90^\circ, 0^\circ, 90^\circ$ with Co-ord. axes, then

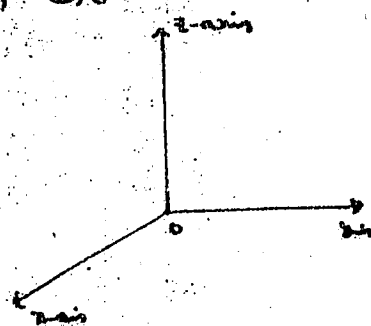
d.c.s. of y-axis are $\cos 90^\circ, \cos 0^\circ, \cos 90^\circ$

$$\text{or } 0, 1, 0$$

Since z-axis makes angles $90^\circ, 90^\circ, 0^\circ$ with Co-ord. axes, so d.c.s. of z-axis are

$$\cos 90^\circ, \cos 90^\circ, \cos 0^\circ$$

$$\text{or } 0, 0, 1$$



Q6 Prove that if, the direction angles of a st. line are α, β & γ then $\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$

Soln.

We know that if direction angles of a line are α, β, γ then d.c.s. are $\cos \alpha, \cos \beta, \cos \gamma$

Also we know

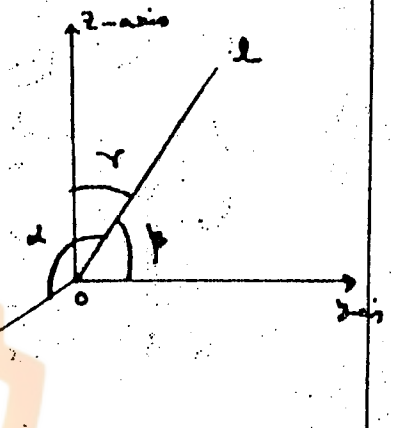
$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

$$3 - (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma) = 1$$

$$-(\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma) = -2$$

$$\Rightarrow \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$



Q7 If measures of two of the direction angles of a st. line are 45° & 60° , find measures of the third direction angle.

Soln.

Let α, β, γ be the direction angles of line

$$\text{then } \alpha = 45^\circ$$

$$\beta = 60^\circ$$

$$\gamma = ?$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\cos^2 45^\circ + \cos^2 60^\circ + \cos^2 \gamma = 1$$

$$\frac{1}{2} + \frac{1}{4} + \cos^2 \gamma = 1$$

$$\cos^2 \gamma = 1 - \frac{1}{2} - \frac{1}{4}$$

$$= \frac{4-2-1}{4}$$

$$= \frac{1}{4}$$

$$\cos^2 \gamma = \frac{1}{4}$$

$$\cos \gamma = \frac{1}{2}$$

$$\boxed{\gamma = 60^\circ}$$

Q8 The d.c.s. of two st. lines are given by the
eqs. $l+m+n=0$, $l^2+m^2-n^2=0$
Find the measure of the angle b/w them.

Soln

Direction Cosines of two lines are given by

$$l+m+n=0 \quad \text{--- (1)}$$

$$l^2+m^2-n^2=0 \quad \text{--- (2)}$$

$$\text{From (1) } n = -(l+m) \quad \text{--- (3)}$$

Put in (2)

$$l^2+m^2-[-(l+m)]^2=0$$

$$l^2 + m^2 - (l^2 + m^2 + 2lm) = 0$$

$$l^2 + m^2 - l^2 - m^2 - 2lm = 0$$

$$-2lm = 0$$

$$lm = 0$$

$$l = 0$$

Put in ③

$$n = -m$$

$$\text{or } \frac{n}{1} = \frac{m}{-1}$$

$$\text{So } \frac{l}{0} = \frac{m}{-1} = \frac{n}{1} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow l_1 = 0$$

$$m_1 = -\frac{1}{\sqrt{2}}$$

$$n_1 = \frac{1}{\sqrt{2}}$$

are req. d. c.

$$m = 0$$

Put in ③

$$n = -l$$

$$\frac{n}{1} = \frac{l}{-1}$$

$$\text{So } \frac{l}{-1} = \frac{m}{0} = \frac{n}{1} = \frac{1}{\sqrt{2}}$$

$$\Rightarrow l_2 = -\frac{1}{\sqrt{2}}$$

$$m_2 = 0$$

$$n_2 = \frac{1}{\sqrt{2}}$$

are req. d. c.

Let θ be the angle b/w given lines then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= (0)\left(-\frac{1}{\sqrt{2}}\right) + \left(-\frac{1}{\sqrt{2}}\right)(0) + \left(\frac{1}{\sqrt{2}}\right)\left(\frac{1}{\sqrt{2}}\right)$$

$$= 0 + 0 + \frac{1}{2}$$

$$\cos \theta = \frac{1}{2}$$

$$\Rightarrow \boxed{\theta = 60^\circ}$$

Q9 The direction Cosines of two str. lines are given by the eqs. $l+m+n=0$ & $2lm+2ln-mn=0$. Find the measure of the angle b/w them.

Sol:- The d.c.s. of given lines are given by eqs.

$$l+m+n=0 \quad \text{--- (1)}$$

$$2lm+2ln-mn=0 \quad \text{--- (2)}$$

from (1) $n = -(l+m) \quad \text{--- (3)}$

Put in (2)

$$2lm+2l(-l-m)-m(-l-m)=0$$

$$2lm-2l(l+m)+m(l+m)=0$$

$$2lm-2l^2-2lm+lm+m^2=0$$

$$-2l^2+lm+m^2=0$$

$$2l^2-lm-m^2=0$$

$$2l^2-2lm+lm-m^2=0$$

$$2l(l-m)+m(l-m)=0$$

$$(l-m)(2l+m)=0$$

$$l-m=0$$

$$\text{or } l=m \Rightarrow \frac{l}{1} = \frac{m}{1}$$

Put in (3)

$$n = -2l$$

$$\frac{n}{-2} = \frac{l}{1}$$

$$2l+m=0$$

$$\text{or } m = -2l \Rightarrow \frac{l}{1} = \frac{m}{-2}$$

Put in (3)

$$n = l$$

$$\text{or } \frac{l}{1} = \frac{n}{1}$$

$$\text{s. } \frac{l}{1} = \frac{m}{1} = \frac{n}{-2} = \frac{1}{\sqrt{6}}$$

$$\Rightarrow \left. \begin{aligned} l_1 &= \frac{1}{\sqrt{6}} \\ m_1 &= \frac{1}{\sqrt{6}} \\ n_1 &= -\frac{2}{\sqrt{6}} \end{aligned} \right\}$$

are d.c.s. for l_1

$$\text{or } \frac{l}{1} = \frac{m}{-2} = \frac{n}{1} = \frac{1}{\sqrt{6}}^{25}$$

$$\Rightarrow \left. \begin{aligned} l_2 &= \frac{1}{\sqrt{6}} \\ m_2 &= -\frac{2}{\sqrt{6}} \\ n_2 &= \frac{1}{\sqrt{6}} \end{aligned} \right\}$$

are d.c.s. for l_2

let θ be the angle b/w given lines then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

$$= \left(\frac{1}{\sqrt{6}}\right)\left(\frac{1}{\sqrt{6}}\right) + \left(\frac{1}{\sqrt{6}}\right)\left(-\frac{2}{\sqrt{6}}\right) + \left(-\frac{2}{\sqrt{6}}\right)\left(\frac{1}{\sqrt{6}}\right)$$

$$= \frac{1}{6} - \frac{2}{6} - \frac{2}{6}$$

$$= \frac{1-2-2}{6}$$

$$= -\frac{3}{6}$$

$$\cos \theta = -\frac{1}{2}$$

$$\Rightarrow \theta = \cos^{-1}\left(-\frac{1}{2}\right)$$

$$\boxed{\theta = 60^\circ}$$

Q10 Find eqs. of the st. lines L & M in symmetric forms. Determine whether the pairs of lines intersect. Find the pt. of intersection if it exists. Prob(10-12)

Q1. L : Through A(2, 1, 3), B(-1, 2, -4)

M : Through P(5, 1, -2), Q(0, 4, 3)

Sol.

The eq. of the line through A(2, 1, 3) & B(-1, 2, -4)

$$\text{is } \frac{x-2}{2-1} = \frac{y-1}{1-2} = \frac{z-3}{3-4}$$

$$\text{or } \frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-3}{-7} \quad \text{--- (L)}$$

Now eq. of line M through P(5, 1, -2) & Q(0, 4, 3) is

$$\frac{x-5}{5-0} = \frac{y-1}{1-4} = \frac{z+2}{-2-3}$$

$$\frac{x-5}{5} = \frac{y-1}{-3} = \frac{z+2}{-5} \quad \text{--- (M)}$$

which are req. eqs. in symmetric form of L & M.

Now we write eqs of L & M in parametric form.

$$\text{let } \frac{x-2}{3} = \frac{y-1}{-1} = \frac{z-3}{-7} = p$$

$$\Rightarrow \left. \begin{aligned} x &= 2+3p \\ y &= 1-p \\ z &= 3-7p \end{aligned} \right\} \quad \text{--- (L)}$$

$$\text{Also let } \frac{x-5}{5} = \frac{y-1}{-3} = \frac{z+2}{-5} = q$$

$$\Rightarrow \left. \begin{aligned} x &= 5+5q \\ y &= 1-3q \\ z &= -2-5q \end{aligned} \right\} \quad \text{--- (M)}$$

Let the lines L & M intersect at (x_0, y_0, z_0)

So this pt. will lie on both lines.

$$\begin{array}{l} \text{So } \left. \begin{array}{l} x_0 = 2+3p \\ y_0 = 1-p \\ z_0 = 3+7p \end{array} \right\} + \left. \begin{array}{l} x_0 = 5+5q \\ y_0 = 1-3q \\ z_0 = -2-5q \end{array} \right\} \end{array}$$

$$\Rightarrow \begin{array}{l} 2+3p = 5+5q \\ 1-p = 1-3q \\ 3+7p = -2-5q \end{array}$$

$$\text{or } 3p-5q = 3 \quad \text{--- (1)}$$

$$p-3q = 0 \quad \text{--- (2)}$$

$$7p+5q = -5 \quad \text{--- (3)}$$

Multiplying (2) by 3

$$3p-5q = 3 \quad \text{--- (1)}$$

$$-3p+9q = 0 \quad \text{--- (2)}$$

$$4q = 3$$

$$\boxed{q = \frac{3}{4}}$$

Put in (2)

$$p-3\left(\frac{3}{4}\right) = 0$$

$$\boxed{p = \frac{9}{4}}$$

We see these values of p & q do not satisfy eq (3)

Hence given lines do not intersect.

So pt. of intersection of lines does not exist.

Q11 $L: \underline{r} = (3\hat{i} + 2\hat{j} - \hat{k}) + t(6\hat{i} - 4\hat{j} - 3\hat{k})$

$M: \underline{r} = (5\hat{i} + 4\hat{j} + 7\hat{k}) + s(14\hat{i} - 6\hat{j} + 2\hat{k})$

Sol: Given lines are

$L: \underline{r} = (3\hat{i} + 2\hat{j} - \hat{k}) + t(6\hat{i} - 4\hat{j} - 3\hat{k})$

$M: \underline{r} = (5\hat{i} + 4\hat{j} + 7\hat{k}) + s(14\hat{i} - 6\hat{j} + 2\hat{k})$

$L: \underline{r} = (3+6t)\hat{i} + (2-4t)\hat{j} + (-1-3t)\hat{k}$

$M: \underline{r} = (5+14s)\hat{i} + (4-6s)\hat{j} + (7+2s)\hat{k}$

Now parametric eq. for L are

$$\left. \begin{aligned} x &= 3+6t \\ y &= 2-4t \\ z &= -1-3t \end{aligned} \right\} \text{--- (L)}$$

& parametric eq. for line M are

$$\left. \begin{aligned} x &= 5+14s \\ y &= 4-6s \\ z &= 7+2s \end{aligned} \right\} \text{--- (M)}$$

Now eq. of L & M in symmetric form are

$$\frac{x-3}{6} = \frac{y-2}{-4} = \frac{z+1}{-3} \text{--- (L)}$$

$$\& \frac{x-5}{14} = \frac{y-4}{-6} = \frac{z-7}{2} \text{--- (M)}$$

Let the lines L & M intersect at pt. (x_0, y_0, z_0)

So (x_0, y_0, z_0) will lie on both lines L & M

Hence it satisfies both eqs.

$$\text{So } \left. \begin{aligned} x_0 &= 3+6t \\ y_0 &= 2-4t \\ z_0 &= -1-3t \end{aligned} \right\} \quad \& \quad \left. \begin{aligned} x_0 &= 5+14s \\ y_0 &= 4-6s \\ z_0 &= 7+2s \end{aligned} \right\} \quad 29$$

Hence

$$\left. \begin{aligned} 3+6t &= 5+14s \\ 2-4t &= 4-6s \\ -1-3t &= 7+2s \end{aligned} \right\}$$

$$\text{or } 6t - 14s = 2$$

$$3t - 7s = 1 \quad \text{--- (1)}$$

$$4t - 6s = -2 \quad \text{--- (2)}$$

$$3t + 2s = -8 \quad \text{--- (3)}$$

$$\text{Subst. (1) \& (3)}$$

$$-9s = 9 \Rightarrow \boxed{s = -1}$$

Put in (1)

$$3t - 7(-1) = 1$$

$$3t = 1-7$$

$$3t = -6$$

$$\Rightarrow \boxed{t = -2}$$

These values of t & s satisfy eq. (2)

Hence lines L & M intersect.

Put $t = -2$ in above eq.

$$\begin{aligned} (x_0, y_0, z_0) &= (3+6(-2), 2-4(-2), -1-3(-2)) \\ &= (-9, 10, 5) \end{aligned}$$

Find the distance of the given pt. P from the given line L: (Problems 13-14).

32

Q13 $P = (3, -2, 1)$ $L: \begin{cases} x = 1+t \\ y = 3-2t \\ z = -2+2t \end{cases}$

Sol: Given pt. & given line are

$P = (3, -2, 1)$ $L: \begin{cases} x = 1+t \\ y = 3-2t \\ z = -2+2t \end{cases}$

A point on the line L is $A = (1, 3, -2)$

Now $\vec{AP} = [3-1, -2-3, 1+2]$
 $= [2, -5, 3]$

now direction vector of the line L is $\underline{b} = [1, -2, 2]$

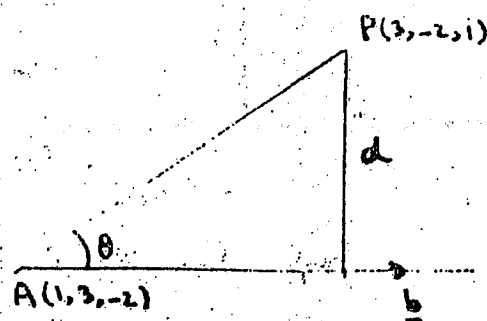
let d be the req. distance of pt. from line
 then

$d = \frac{|\vec{AP} \times \underline{b}|}{|\underline{b}|} \quad \text{--- (1)}$

Now $\vec{AP} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & -5 & 3 \\ 1 & -2 & 2 \end{vmatrix}$

$= \hat{i}(-10+6) - \hat{j}(4-3) + \hat{k}(-4+5)$

$= -4\hat{i} - \hat{j} + \hat{k}$



$$\begin{aligned} \therefore |\vec{AP} \times \underline{b}| &= \sqrt{16+1+1} \\ &= \sqrt{18} \\ &= 3\sqrt{2} \end{aligned}$$

$$\begin{aligned} \text{Now } |\underline{b}| &= \sqrt{1+4+4} \\ &= \sqrt{9} \\ &= 3 \end{aligned}$$

Put in ①

$$d = \frac{3\sqrt{2}}{3}$$

$$\boxed{d = \sqrt{2}}$$

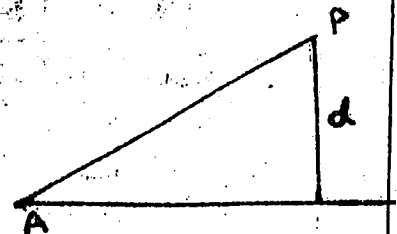
Q14 $P = (0, -2, 1)$, $L: \frac{x-1}{4} = \frac{y+3}{-2} = \frac{z+1}{5}$

Sol: Given pt. & line are

$$P = (0, -2, 1); \quad L: \frac{x-1}{4} = \frac{y+3}{-2} = \frac{z+1}{5}$$

A pt. on the line L is $A = (1, -3, -1)$

$$\begin{aligned} \text{Now } \vec{AP} &= [0-1, -2+3, 1+1] \\ &= [-1, 1, 2] \end{aligned}$$

Now direction vector of line L is $\underline{b} = [4, -2, 5]$

Let d be the req. distance then

$$d = \frac{|\vec{AP} \times \underline{b}|}{|\underline{b}|} \quad \text{--- ①}$$

$$\text{Now } \vec{AP} \times \underline{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -1 & 1 & 2 \\ 4 & -2 & 5 \end{vmatrix}$$

$$= \hat{i}(5+4) - \hat{j}(-5-8) + \hat{k}(2-4)$$

$$= 9\hat{i} + 13\hat{j} - 2\hat{k}$$

$$+ |\vec{AP} \times \underline{b}| = \sqrt{81 + 169 + 4}$$

$$= \sqrt{254}$$

$$\text{Now } |\underline{b}| = \sqrt{16 + 4 + 25}$$

$$= \sqrt{45}$$

Put in ①

$$d = \frac{\sqrt{254}}{\sqrt{45}} \quad \text{Ans.}$$

Q15 If the edges of a rectangular parallelepiped are a, b, c , show that the angles b/w the four diagonals are given by

$$\cos^{-1} \left(\frac{a^2 + b^2 + c^2}{a^2 + b^2 + c^2} \right)$$

Sol. Here the lengths of the edges OA, OB & OC are a, b, c resp.

The Co-ords. of the vertices of parallelepiped are $O = (0, 0, 0)$, $O' = (a, b, c)$, $A = (a, 0, 0)$, $A' = (0, b, c)$

$$B = (0, b, 0), B' = (a, 0, c), C = (0, 0, c) \text{ \& } C' = (a, b, 0) \text{ is}$$

The four diagonals of parallelepiped are

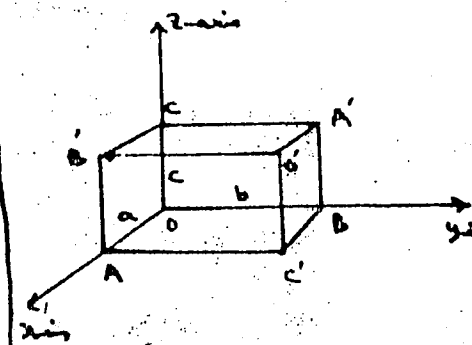
$$AA', BB', CC' \text{ \& } OO'$$

D.s. of AA' are $-a, b, c$

D.s. of BB' are $a, -b, c$

D.s. of CC' are $a, b, -c$

D.s. of OO' are a, b, c



So d.s. of AA' are $\frac{-a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$

d.s. of BB' are $\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{-b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$

d.s. of CC' are $\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{-c}{\sqrt{a^2+b^2+c^2}}$

d.s. of OO' are $\frac{a}{\sqrt{a^2+b^2+c^2}}, \frac{b}{\sqrt{a^2+b^2+c^2}}, \frac{c}{\sqrt{a^2+b^2+c^2}}$

Angle b/w diagonal AA' \& BB' is

$$\cos \theta = \frac{-a^2}{a^2+b^2+c^2} + \frac{-b^2}{a^2+b^2+c^2} + \frac{c^2}{a^2+b^2+c^2}$$

$$\cos \theta = \frac{-a^2-b^2+c^2}{a^2+b^2+c^2}$$

The angle b/w diagonal BB' & CC' is

$$\cos \beta = \frac{a^2}{a^2+b^2+c^2} + \frac{-b^2}{a^2+b^2+c^2} + \frac{-c^2}{a^2+b^2+c^2}$$

36

$$\cos \beta = \frac{a^2 - b^2 - c^2}{a^2 + b^2 + c^2}$$

$$\Rightarrow \beta = \cos^{-1} \left(\frac{a^2 - b^2 - c^2}{a^2 + b^2 + c^2} \right) \quad \text{--- (2)}$$

The angle b/w diagonals CC' & OO' is

$$\cos \gamma = \frac{a^2}{a^2+b^2+c^2} + \frac{b^2}{a^2+b^2+c^2} + \frac{-c^2}{a^2+b^2+c^2}$$

$$\cos \gamma = \frac{a^2 + b^2 - c^2}{a^2 + b^2 + c^2}$$

$$\Rightarrow \gamma = \cos^{-1} \left(\frac{a^2 + b^2 - c^2}{a^2 + b^2 + c^2} \right) \quad \text{--- (3)}$$

The angle b/w the diagonal AA' & OO' is

$$\cos \delta = \frac{-a^2}{a^2+b^2+c^2} + \frac{b^2}{a^2+b^2+c^2} + \frac{c^2}{a^2+b^2+c^2}$$

$$\cos \delta = \frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2}$$

$$\Rightarrow \delta = \cos^{-1} \left(\frac{-a^2 + b^2 + c^2}{a^2 + b^2 + c^2} \right) \quad \text{--- (4)}$$

from (1), (2), (3) & (4), we see that the angles b/w four diagonals of a cube are

$$\cos^{-1} \left(\frac{\pm a^2 \pm b^2 \pm c^2}{a^2 + b^2 + c^2} \right)$$

Q16 A str. line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube. Prove that

37

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

Sol. Consider a cube & let a be the length of one edge of cube.

Then the Co-ords. of different vertices of cube

$$O = (0, 0, 0), O' = (a, a, a)$$

$$A = (a, 0, 0), A' = (0, a, a)$$

$$B = (0, a, 0), B' = (a, 0, a)$$

$$C = (0, 0, a), C' = (a, a, 0)$$

Then AA', BB', CC' & OO' are four diagonals of the cube.

d.s. of diagonal AA' are $-a, a, a$

d.s. of diagonal BB' are $a, -a, a$

d.s. of diagonal CC' are $a, a, -a$

d.s. of diagonal OO' are a, a, a

$$\text{Now } \sqrt{a^2 + a^2 + a^2} = \sqrt{3a^2} = \sqrt{3}a$$

So

d.s. of diagonal AA' are $\frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

d.s. of diagonal BB' are $\frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

d.s. of diagonal CC' are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{-1}{\sqrt{3}}$

& d.s. of diagonal OO' are $\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}$

Since $L \parallel AB$

So d.r.s. of line L are $-1, 3, -2$

Hence eq. of line L passing through $P(0, -3, 2)$ & having d.r.s. $-1, 3, -2$ is

$$\frac{x-0}{-1} = \frac{y+3}{3} = \frac{z+1}{-2}$$

$$\text{or } \frac{x}{1} = \frac{y+3}{-2} = \frac{z+1}{2}$$

Q18 Find, eq. of the st. line passing through the pt. $P(2, 0, -2)$ & perpendicular to each of the st. lines.

$$\frac{x-3}{2} = \frac{y}{2} = \frac{z+1}{2} \quad \text{and} \quad \frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2}$$

Soln.

Given lines are

$$\frac{x-3}{2} = \frac{y}{2} = \frac{z+1}{2} \quad \text{--- (L}_1\text{)}$$

$$\frac{x}{3} = \frac{y+1}{-1} = \frac{z+2}{2} \quad \text{--- (L}_2\text{)}$$

D.r.s. of L_1 are $2, 2, 2$

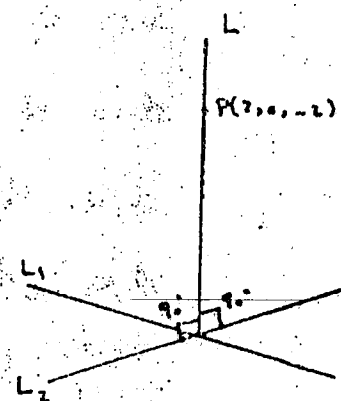
D.r.s. of L_2 are $3, -1, 2$

Suppose L be the req. line with d.r.s. c_1, c_2, c_3

Since $L \perp L_1$ & $L \perp L_2$

So by condition of perpendicularity,

$$\left. \begin{aligned} 2c_1 + 2c_2 + 2c_3 &= 0 \\ \& \quad 3c_1 - c_2 + 2c_3 &= 0 \end{aligned} \right\}$$



Let l, m, n be the d.s. of the line which makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals AA', BB', CC', DD' then

$$\cos \alpha = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} + \frac{n}{\sqrt{3}} = \frac{l+m+n}{\sqrt{3}} \quad \text{--- (1)}$$

$$\cos \beta = \frac{l}{\sqrt{3}} + \frac{-m}{\sqrt{3}} + \frac{n}{\sqrt{3}} = \frac{l-m+n}{\sqrt{3}} \quad \text{--- (2)}$$

$$\cos \gamma = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} - \frac{n}{\sqrt{3}} = \frac{l+m-n}{\sqrt{3}} \quad \text{--- (3)}$$

$$\cos \delta = \frac{l}{\sqrt{3}} + \frac{m}{\sqrt{3}} - \frac{n}{\sqrt{3}} = \frac{l+m-n}{\sqrt{3}} \quad \text{--- (4)}$$

Sq. (1), (2), (3) & (4) & adding.

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4l^2 + 4m^2 + 4n^2}{3}$$

$$= \frac{4(l^2 + m^2 + n^2)}{3}$$

Available at
www.mathcity.org

$$= \frac{4}{3}$$

Q17 Find the eqn. of the str. line passing through the pt. $P(0, -3, 2)$ & parallel to the str. line joining the pts. $A(3, 4, 7)$ & $B(2, 7, 5)$.

Sol.

Let L be the req. line passing through $P(0, -3, 2)$

Now d.s. of line AB are $2-3, 7-4, 5-7$

$$\text{or } -1, 3, -2$$

$$\frac{C_1}{\begin{vmatrix} 2 & 2 \\ -1 & 2 \end{vmatrix}} = \frac{-C_2}{\begin{vmatrix} 2 & 2 \\ 3 & 2 \end{vmatrix}} = \frac{C_3}{\begin{vmatrix} 2 & 2 \\ 3 & -1 \end{vmatrix}}$$

$$\frac{C_1}{4+2} = \frac{-C_2}{4-6} = \frac{C_3}{-2-6}$$

$$\frac{C_1}{6} = \frac{C_2}{2} = \frac{C_3}{-8}$$

$$\frac{C_1}{3} = \frac{C_2}{1} = \frac{C_3}{-4}$$

So d.r.s. of line L are 3, 1, -4

Hence eq. of line L is

$$\frac{x-2}{3} = \frac{y-0}{1} = \frac{z+2}{-4}$$

Find eq. of the str. line through the given pt A & perpendicular to the given str. line (Problems 19-20):

Q19 A = (11, 4, -6) & $x = 4-t$, $y = 7+2t$, $z = -1+t$

Sol: Given pt. & line are

A = (11, 4, -6) & $x = 4-t$, $y = 7+2t$, $z = -1+t$

Let L be the req. line passing through A(11, 4, -6) & perp. to given line. Suppose it meets the given line at pt. B.

Now any pt. on given line is $(4-t, 7+2t, -1+t)$

D.r.s. of AB are $4-t-11$, $7+2t-4$, $-1+t+6$
S. Co. of B are $(4-t, 7+2t, -1+t)$

or $-t-7, 2t+3, t+5$

Here d.r.s. of given line are $-1, 2, 1$

Since $AB \perp$ given line

So $(-1)(-t-7) + 2(2t+3) + 1(t+5) = 0$

$$t+7+4t+6+t+5=0$$

$$6t+18=0$$

$$t+3=0$$

$$t = -3$$

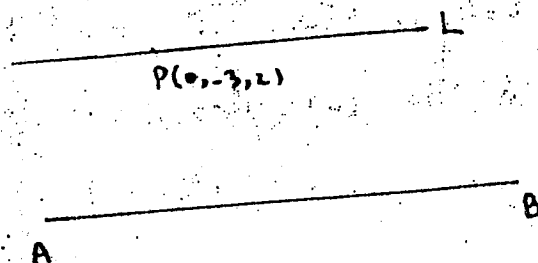
Hence d.r.s. of AB are $3-7, 2(-3)+3, -3+5$
 $= -4, -3, 2$

So eq. of line AB is

$$\frac{x-11}{-4} = \frac{y-1}{-3} = \frac{z+1}{2}$$

or $\frac{x-11}{4} = \frac{y-1}{3} = \frac{z+1}{-2}$

P3



Q20 $A = (5, -4, 4)$ & $\frac{x}{-1} = \frac{y-1}{1} = \frac{z}{-2} = t$

Soln- Given pt. & line is

$$A = (5, -4, 4) \text{ & } \frac{x}{-1} = \frac{y-1}{1} = \frac{z}{-2} = t$$

The parametric eqs. of given line are

$$\left. \begin{aligned} x &= -t \\ y &= 1+t \\ z &= -2t \end{aligned} \right\}$$

Let L be the req. line passing through $A(5, -4, 4)$ & perp. to given line. Suppose it meets the given line at pt. B

Any pt. on given line are $(-t, 1+t, -2t)$

So Co-ords. of pt. B are $B(-t, 1+t, -2t)$

$$\begin{aligned} \text{Dirs. of line AB are } & 5+t, -4-1-t, 4+2t \\ & = 5+t, -5-t, 4+2t \end{aligned}$$

Here dirs. of given line are $-1, 1, -2$

Since $AB \perp$ given line

$$\text{So } -1(5+t) + 1(-5-t) + (-2)(4+2t) = 0$$

$$-5-t - 5-t - 8 - 4t = 0$$

$$-6t - 18 = 0$$

$$t + 3 = 0 \Rightarrow |t| = -3$$

1282

So dir. of AB are $5-3, -5+3, 4-6$

or $2, -2, -2$

or $1, -1, -1$

Hence req. eq. of line AB is

$$\frac{x-5}{1} = \frac{y+4}{-1} = \frac{z-4}{-1}$$

Asghar Mal

Q21 Find the length of the perpendicular from the pt. (x_1, y_1, z_1) to the st. line $\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-r}{n}$

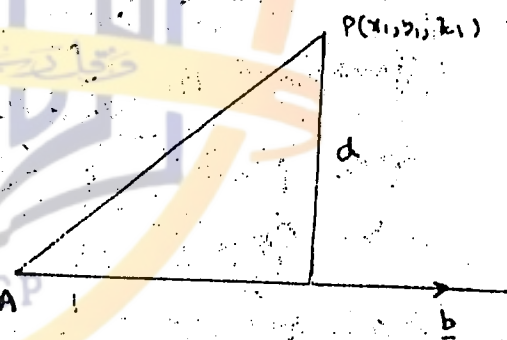
where $l^2 + m^2 + n^2 = 1$

Sol.

Given pt. & given line are

$P(x_1, y_1, z_1)$ &

$$\frac{x-a}{l} = \frac{y-b}{m} = \frac{z-r}{n}$$



Here $A = (a, b, r)$ is

a pt. of given line

Since l, m, n are dir. of given line.

So direction vector of given line is $\underline{b} = [l, m, n]$

Now $\overrightarrow{AP} = [x_1 - a, y_1 - b, z_1 - r]$

Let d be the req. distance then

$$d = \frac{|\vec{AP} \times \vec{b}|}{|\vec{b}|} \quad \text{--- (1)}$$

$$\text{Now } \vec{AP} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x_1 - a & y_1 - b & z_1 - c \\ l & m & n \end{vmatrix}$$

$$= \hat{i} [n(y_1 - b) - m(z_1 - c)] - \hat{j} [n(x_1 - a) - l(z_1 - c)] + \hat{k} [m(x_1 - a) - l(y_1 - b)]$$

$$\vec{AP} \times \vec{b} = [n(y_1 - b) - m(z_1 - c)] \hat{i} + [l(z_1 - c) - n(x_1 - a)] \hat{j} + [m(x_1 - a) - l(y_1 - b)] \hat{k}$$

$$\Rightarrow |\vec{AP} \times \vec{b}| = \sqrt{[n(y_1 - b) - m(z_1 - c)]^2 + [l(z_1 - c) - n(x_1 - a)]^2 + [m(x_1 - a) - l(y_1 - b)]^2}$$

$$= \sqrt{\sum [n(y_1 - b) - m(z_1 - c)]^2}$$

Put in (1)

$$d = \frac{\sqrt{\sum [n(y_1 - b) - m(z_1 - c)]^2}}{\sqrt{l^2 + m^2 + n^2}}$$

$$d = \sqrt{\sum [n(y_1 - b) - m(z_1 - c)]^2}$$

is the req. distance.

پروفیسر ایڈم سٹیشنرز
نزد گورنمنٹ پوسٹ گریجویٹ کالج اصفہان
مادریںڈی - قوا

Q22 Find eq. of the perpendicular from the pt. $(1, 6, 3)$ to the st. line

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

Also obtain its length & Co-ords. of the foot of the perpendicular.

Sol.

Given pt. & given line are $A = (1, 6, 3)$ &

$$\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$$

Let AP be the length of perpendicular from pt. A to given line.

$$\therefore \frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3} = t$$

So parametric eqs. of given line are

$$\left. \begin{aligned} x &= t \\ y &= 1+2t \\ z &= 2+3t \end{aligned} \right\}$$

Any pt. on this line is $(t, 1+2t, 2+3t)$

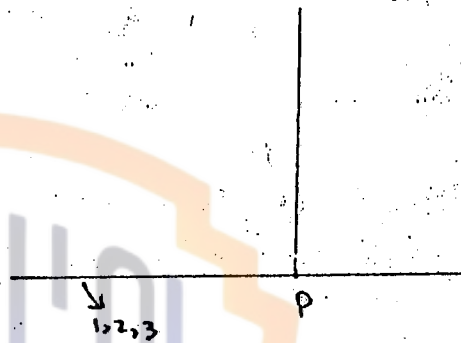
So Co-ords. of pt. P are $P(t, 1+2t, 2+3t)$

Dirs. of line AP are $t-1, 1+2t-6, 2+3t-3$

$$= t-1, 2t-5, 3t-1$$

& dirs. of given line are $1, 2, 3$.

A(1,6,3)



Since AP $\perp$ given line

$$\text{So } 1(t-1) + 2(2t-5) + 3(3t-1) = 0$$

$$t-1 + 4t-10 + 9t-3 = 0$$

$$14t - 14 = 0$$

$$t-1 = 0$$

$$|t=1|$$

So Co-ords. of pt. P are $P(1, 3, 5)$.

$$\text{Length of perp.} = |AP|$$

$$= \sqrt{(1-1)^2 + (3-1)^2 + (5-3)^2}$$

$$= \sqrt{0 + 4 + 4}$$

$$= \sqrt{8}$$

Now eq. of perpendicular AP is

$$\frac{x-1}{1-1} = \frac{y-6}{3-6} = \frac{z-3}{5-3}$$

$$\text{or } \frac{x-1}{0} = \frac{y-6}{-3} = \frac{z-3}{2}$$

Q23 Find a necessary & sufficient condition that the pts. $P(x_1, y_1, z_1)$, $Q(x_2, y_2, z_2)$ & $R(x_3, y_3, z_3)$ are collinear.

Sol. Given pts. are

$$P(x_1, y_1, z_1), Q(x_2, y_2, z_2) \text{ \& } R(x_3, y_3, z_3)$$

Suppose that the pts. P, Q & R are collinear.

Now eq. of line through $P(x_1, y_1, z_1)$ & $Q(x_2, y_2, z_2)$ is

$$\frac{x-x_1}{x_2-x_1} = \frac{y-y_1}{y_2-y_1} = \frac{z-z_1}{z_2-z_1}$$

Since pts. P, Q & R are collinear

So $R(x_3, y_3, z_3)$ lies on above line

$$\therefore \frac{x_3-x_1}{x_2-x_1} = \frac{y_3-y_1}{y_2-y_1} = \frac{z_3-z_1}{z_2-z_1} = t \text{ (say)}$$

$$\Rightarrow x_3 - x_1 = t(x_2 - x_1)$$

$$y_3 - y_1 = t(y_2 - y_1)$$

$$z_3 - z_1 = t(z_2 - z_1)$$

$$\text{or } x_3 - x_1 - tx_2 + tx_1 = 0$$

$$y_3 - y_1 - ty_2 + ty_1 = 0$$

$$z_3 - z_1 - tz_2 + tz_1 = 0$$

$$\text{or } (t-1)x_1 - tx_2 + x_3 = 0$$

$$(t-1)y_1 - ty_2 + y_3 = 0$$

$$(t-1)z_1 - tz_2 + z_3 = 0$$

Eliminating t from above eqs.

$$\begin{vmatrix} x_1 & x_2 & x_3 \\ y_1 & y_2 & y_3 \\ z_1 & z_2 & z_3 \end{vmatrix} = 0$$

$$\text{or } \begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$

which is a necessary condition for three pts. P, Q & R to be collinear.

Conversely Suppose that for three pts. $P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$ & $R(x_3, y_3, z_3)$

$$\begin{vmatrix} x_1 & y_1 & z_1 \\ x_2 & y_2 & z_2 \\ x_3 & y_3 & z_3 \end{vmatrix} = 0$$



Q24 If l_1, m_1, n_1 ; l_2, m_2, n_2 ; l_3, m_3, n_3 are d.c.s. of three mutually perpendicular lines, prove that the line whose d.c.s. are proportional to $l_1 + l_2 + l_3$, $m_1 + m_2 + m_3$ & $n_1 + n_2 + n_3$ makes congruent angles with them.

Sol:-

Suppose that L_1, L_2 & L_3 are given lines s.t.

d.c.s. of L_1 are l_1, m_1, n_1

d.c.s. of L_2 are l_2, m_2, n_2

d.c.s. of L_3 are l_3, m_3, n_3

Since lines L_1, L_2, L_3 are mutually perpendicular

$$\left. \begin{aligned} l_1 l_2 + m_1 m_2 + n_1 n_2 &= 0 \\ l_2 l_3 + m_2 m_3 + n_2 n_3 &= 0 \\ l_1 l_3 + m_1 m_3 + n_1 n_3 &= 0 \end{aligned} \right\}$$

Let L be the line having d.c.s. $l_1 + l_2 + l_3$, $m_1 + m_2 + m_3$ & $n_1 + n_2 + n_3$

Let θ_1 be the angle b/w L & L_1 then

$$\begin{aligned} \cos \theta_1 &= \frac{l_1(l_1 + l_2 + l_3) + m_1(m_1 + m_2 + m_3) + n_1(n_1 + n_2 + n_3)}{\sqrt{l_1^2 + m_1^2 + n_1^2} \cdot \sqrt{(l_1 + l_2 + l_3)^2 + (m_1 + m_2 + m_3)^2 + (n_1 + n_2 + n_3)^2}} \\ &= \frac{(l_1^2 + m_1^2 + n_1^2) + (l_1 l_2 + m_1 m_2 + n_1 n_2) + (l_1 l_3 + m_1 m_3 + n_1 n_3)}{\sqrt{(l_1^2 + m_1^2 + n_1^2) + (l_2^2 + m_2^2 + n_2^2) + (l_3^2 + m_3^2 + n_3^2) + 2(l_1 l_2 + m_1 m_2 + n_1 n_2) + 2(l_2 l_3 + m_2 m_3 + n_2 n_3) + 2(l_1 l_3 + m_1 m_3 + n_1 n_3)}} \\ &= \frac{1 + 0 + 0}{\sqrt{1 + 1 + 1 + 0 + 0 + 0}} \end{aligned}$$

$$\cos \theta_1 = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta_1 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ ————— ①}$$

Let θ_2 be the angle b/w L & L_2 then

$$\begin{aligned} \cos \theta_2 &= \frac{l_2(l_1+l_2+l_3) + m_2(m_1+m_2+m_3) + n_2(n_1+n_2+n_3)}{\sqrt{l_2^2+m_2^2+n_2^2} \cdot \sqrt{(l_1+l_2+l_3)^2 + (m_1+m_2+m_3)^2 + (n_1+n_2+n_3)^2}} \\ &= \frac{(l_1l_2 + m_1m_2 + n_1n_2) + (l_2^2 + m_2^2 + n_2^2) + (l_2l_3 + m_2m_3 + n_2n_3)}{\sqrt{(l_1^2+m_1^2+n_1^2) + (l_2^2+m_2^2+n_2^2) + (l_3^2+m_3^2+n_3^2)} + 2(l_1l_2 + m_1m_2 + n_1n_2) + 2(l_1l_3 + m_1m_3 + n_1n_3) + 2(l_2l_3 + m_2m_3 + n_2n_3)} \\ &= \frac{0+1+0}{\sqrt{1+1+1+0+0+0}} \end{aligned}$$

$$\cos \theta_2 = \frac{1}{\sqrt{3}}$$

$$\Rightarrow \theta_2 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ ————— ②}$$

Now let θ_3 be the angle b/w L & L_3 then

$$\begin{aligned} \cos \theta_3 &= \frac{l_3(l_1+l_2+l_3) + m_3(m_1+m_2+m_3) + n_3(n_1+n_2+n_3)}{\sqrt{l_3^2+m_3^2+n_3^2} \cdot \sqrt{(l_1+l_2+l_3)^2 + (m_1+m_2+m_3)^2 + (n_1+n_2+n_3)^2}} \\ &= \frac{(l_1l_3 + m_1m_3 + n_1n_3) + (l_2l_3 + m_2m_3 + n_2n_3) + (l_3^2 + m_3^2 + n_3^2)}{1 \cdot \sqrt{3}} \end{aligned}$$

$$= \frac{0+0+1}{\sqrt{3}}$$

$$\cos \theta_3 = \frac{1}{\sqrt{3}}$$

$$\theta_3 = \cos^{-1}\left(\frac{1}{\sqrt{3}}\right) \text{ ————— ③}$$

from ①, ② & ③ it is proved that

$$\theta_1 = \theta_2 = \theta_3 \text{ ————— Proved}$$

Q25 A variable line in two adjacent positions hasst direction cosines l, m, n & $l+\delta l, m+\delta m$ & $n+\delta n$. Showst that the measure of the small angle $\delta\theta$ b/w the two positions is given by

$$(\delta\theta)^2 = (\delta l)^2 + (\delta m)^2 + (\delta n)^2$$

Sol.

Let OA & OB be the two adjacent positions of the line. Let PQ be the arc of the circle with centre at O & radius 1.

Then the Co-ords. of pts.

P & Q are $P(l, m, n)$ & $Q(l+\delta l, m+\delta m, n+\delta n)$

Let $\delta\theta$ be the angle b/w two positions of line.

Now $\delta\theta = \widehat{PQ} = \text{chord PQ}$ Since $\delta\theta \rightarrow 0$

$$\text{So } \delta\theta = |PQ|$$

$$= \sqrt{(l+\delta l - l)^2 + (m+\delta m - m)^2 + (n+\delta n - n)^2}$$

$$\delta\theta = \sqrt{\delta l^2 + \delta m^2 + \delta n^2}$$

$$\Rightarrow \boxed{\delta\theta^2 = \delta l^2 + \delta m^2 + \delta n^2}$$

